HU LIFAN

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EDUCATION

National University of Singapore | B.Eng. Computer Engineering | Aug 2024 - Present
Second Major: Innovation and Design Program; Minor: Mathematics
Shanghai Jiao Tong University | Summer School | Jun - Jul 2025
Coursework: Algebra, Statistical Inference

RESEARCH EXPERIENCE

DistDNA: Language Model Fingerprinting under Stochastic Decoding | NUS
Research Assistant under Dr. Wu Zhaomin; supervised by Prof. He Bingsheng | Aug 2026 - Present
- Researched black-box LLM identification using prompt-conditional response distributions and maximum mean discrepancy (MMD) under temperature and top-p shifts.
- Studied finite-sample fingerprint estimation and nearest-reference identification without a trained classifier.

Graph Evidence Is Not Text Evidence | Independent Research | Jun 2026 - Present
- Proposed structural invariance as a criterion for evaluating graph-grounded language models beyond task accuracy.
- Developed evaluations for serialization-order sensitivity, graph-isomorphism consistency, format stability, semantic interference, and distractor robustness.

On-Device Vision-Language Assistant for the Visually Impaired | NUS | Aug 2025 - Feb 2026
Undergraduate Researcher; supervised by A/Prof. Lim Li Hong Idris
- Implemented an offline multimodal perception pipeline with quantized VILA1.5-3B and Qwen2.5-VL-3B on NVIDIA Jetson Orin Nano Super for assistive vision.
- Benchmarked latency, memory footprint, and throughput across VLM architectures and quantization schemes to evaluate deployment tradeoffs on embedded GPUs.

Adaptive Graph Evidence Acquisition | Research supervised by Prof. He Bingsheng, NUS
May - Aug 2026
- Developed a graph evidence-acquisition framework separating cost-aware evidence construction from downstream classification and language-model-based fraud reasoning.
- Implemented graph-learning and retrieval baselines, including GCN, GraphSAGE, node2vec, and Personalized PageRank, with multi-seed evaluation and controlled ablations.

ENGINEERING EXPERIENCE

TikTok Pte. Ltd. | Machine Learning Engineer Intern, BRIC | Singapore | May 2026 - Present
- Developed graph-based integrity-risk discovery methods for live-streaming ecosystems using k-hop subgraphs, community detection, Personalized PageRank, and structural anomaly signals.
- Built distributed Spark pipelines to rank accounts, transactions, and coordinated communities for investigation; designed anomaly-scoring methods under incomplete and weak supervision.

Shanghai MAHLE Thermal Systems Co., Ltd. | Digital Developer Intern | May - Jul 2025
- Built an offline multilingual meeting transcriber and translator with Vosk and MarianMT; trained YOLOv5 for manufacturing-document layout detection.
- Implemented OCR pipelines with Tesseract and PaddleOCR, and deployed Flask APIs for inference, OCR, and monitoring dashboards on local networks.

SELECTED PUBLICATIONS

Lifan Hu. "Graph Evidence Is Not Text Evidence: Serialization Faithfulness as Structural Invariance in Graph-Grounded Language Models." **EMNLP 2026 GroundLM Workshop**; accepted archival short paper and poster.

Lifan Hu, Chew Hoe Alfred Ang, Li Hong Idris Lim. "Quantized Vision-language models on Embedded GPUs: A Comparative Benchmark for Edge Multimodal Inference." **ICARCV 2026**; accepted oral presentation.

TECHNICAL SKILLS & RECOGNITION

Languages & systems: Python, C/C++, Java, SQL, PySpark, Git, Linux, Docker, Flask, FastAPI
ML & deployment: PyTorch, PyTorch Geometric, scikit-learn, GNNs, transformers, VLMs, reinforcement learning, OpenCV, TensorRT, NVIDIA Jetson, model quantization
Awards: CDE Innovation & Research Award, Merit (2026); Mathematical Contest in Modeling, Meritorious Winner (2025), Honorable Mention (2026)